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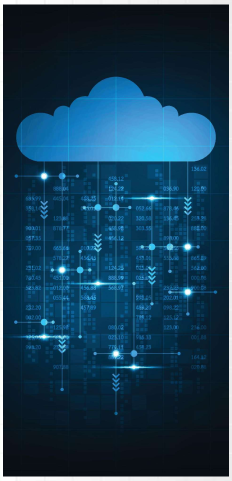

Cloud programming

Rafał Palak

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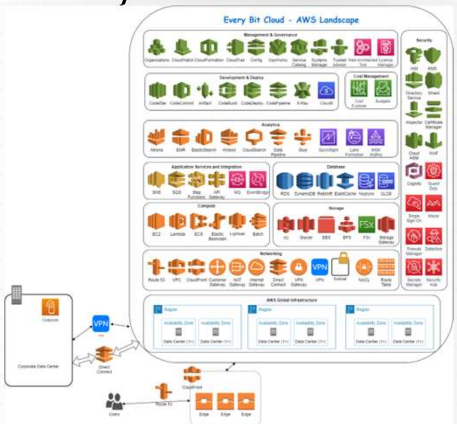
AWS Ecosystem



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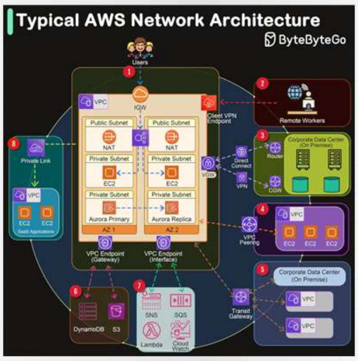
Typical AWS architecture



https://blog.bytebytego.com/p/ep103-typical-aws-network-architecture

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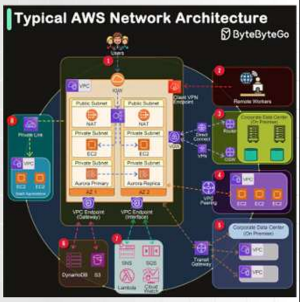
VPC (Virtual Private Cloud)

- It allows you to run AWS resources in a virtual network that can be fully configured according to your own needs.



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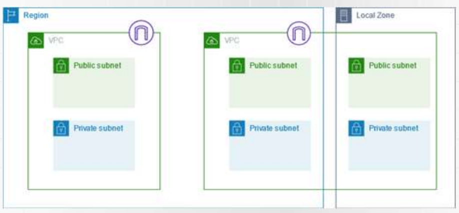
Subnets

- They allow you to divide the VPC into smaller, isolated sections. You can create public and private subnets.



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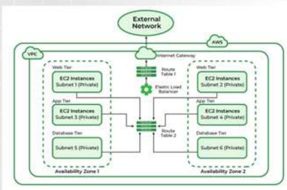
Subnets – why? [1]

- **Network Traffic Segregation** - Subnets allow you to separate network traffic between different groups of resources, which helps in managing access and controlling data flow.
- **Increased Security** - Resources in different subnets can be isolated from each other, preventing unauthorized access and enhancing application security.
- **Logical Grouping of Resources** - Dividing a VPC into subnets allows for the logical grouping of resources based on their function, e.g., application layer, database layer.
- **Easier Management** - With subnets, resource management becomes more transparent and easier to monitor.



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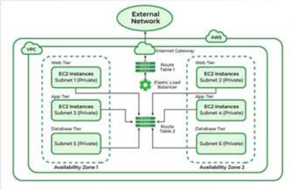
Subnets – why? [2]

- **Public and Private Subnets -** Subnets can be configured as **public (accessible from the Internet)** or **private (internally accessible only)**, allowing for control over resource access.
  - **Public Subnets** - Host resources that need to be accessible from the Internet, such as web servers.
  - **Private Subnets** - Host resources that do not need to be directly accessible from the Internet, such as databases.
- **Performance Optimization** - Segmenting the network into subnets **allows for the optimization of network traffic and resources**, which can **improve application performance**.
- **Better Control Over Network Traffic** - The ability to **create dedicated routing paths for different subnets** allows for better control and optimization of network traffic.



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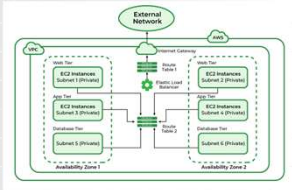
Subnets – why? [3]

- **Compliance with Security Policies -** Subnetting allows you to meet regulatory requirements and security policies through **isolation and access control**.
- **Monitoring and Auditing -** The ability to monitor network traffic and activities across different subnets enables better auditing and compliance with security requirements.



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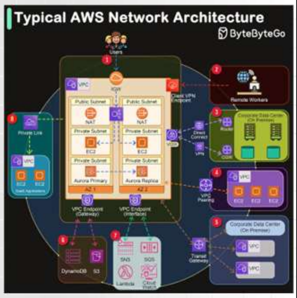
Subnets – Example Scenarios

- **Web Application**:
  - **Public Subnet** - Application servers, load balancers.
  - **Private Subnet** - Databases, back-end application servers that do not require Internet access.
- **Dev/Test Environment**:
  - **Public Subnet** - Testing servers that need to be accessible from outside.
  - **Private Subnet** - Development and testing resources that should be isolated from public traffic.
- **Hybrid Cloud Environment -**
  - **Public Subnet** - VPN gateway for connections to on-premises data centers.
  - **Private Subnet** - Production resources and data that must remain in a secure, private environment.



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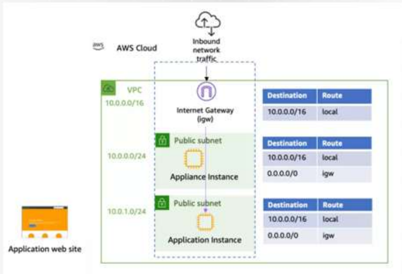
Route Tables

- These are sets of rules (routes) that determine how network traffic is directed within the virtual VPC network.



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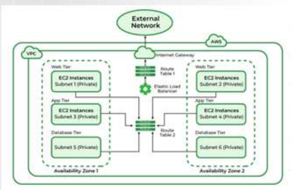
Route Tables – why? [1]

- **Route Rules** - Each route table **contains a set of route rules that determine where to direct network traffic** based on its destination IP address.
- **Default Route Table** - Every VPC has a **default route table** that automatically directs traffic within the VPC.
- **Subnet Associations** - **Each subnet in a VPC must be associated with a route table**. Subnets can have different route tables, allowing for network segmentation and traffic control.
- **Subnet Isolation** - Enables the **isolation of traffic between different subnets**, enhancing security and network control.



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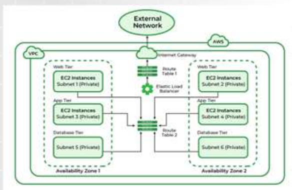
Route Tables – why? [2]

- **Internet Gateway** - Route tables can include routes directing traffic to the Internet via an Internet Gateway.
- **NAT Gateway** - Allows private subnets to access the Internet without exposing their private IP addresses.
- **VPN Gateway and Direct Connect** - Enables connections to on-premises networks through secure VPN connections or dedicated Direct Connect links.
- **VPC Peering -** Route tables can include routes that enable communication between different VPCs within the same or different AWS regions.



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Route Tables – Example Scenarios

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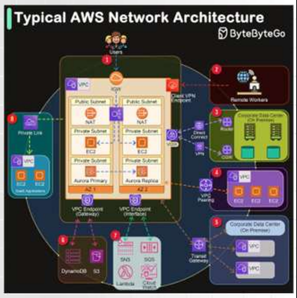

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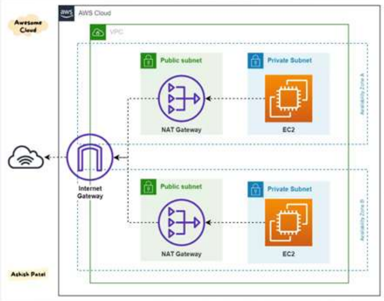
Internet Gateway

- A VPC component that enables communication between resources in the VPC and the Internet.



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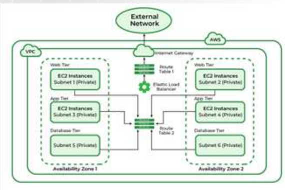
Internet Gateway – why?

- Enables resources in public subnets to send and receive traffic from the Internet.
- Provides Internet access for EC2 instances and other AWS services in public subnets.
- Acts as a gateway that handles routing of Internet traffic to and from the VPC.
- In conjunction with route tables, defines which subnets have Internet access.
- Provides automatic scalability to handle variable traffic loads.
- High availability through redundancy, ensuring reliable Internet connectivity.



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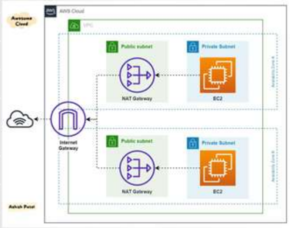
NAT Gateway

- It is a key component of the network infrastructure in Amazon Web Services (AWS). It is used to enable instances in private subnets to access the Internet or other AWS services without exposing their private IP addresses.



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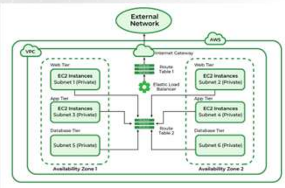
NAT Gateway – why?

- **Instances in private subnets are protected from direct access from the Internet.** Using a NAT Gateway, these instances can initiate outbound connections but cannot receive direct inbound connections from the Internet.
- **NAT Gateway is a managed AWS service, meaning that AWS handles the scaling, management, and maintenance.** Users do not need to configure and manage NAT instances themselves, simplifying infrastructure management.
- **NAT Gateway is automatically scaled and redundant within an Availability Zone**, ensuring high availability and fault tolerance.



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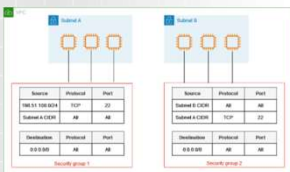
Security Groups

- These are virtual firewalls that control incoming and outgoing traffic to resources running in an Amazon VPC. They are sets of rules that define what traffic is allowed to and from EC2 instances and other resources within the VPC.



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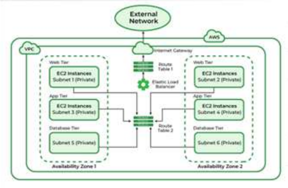
Security Groups – why? [1]

- **Security Groups allow you to define rules that specify what incoming traffic is permitted** to resources assigned to that group.
- Rules can include the **specification of protocols (TCP, UDP, ICMP)**, **ports**, and **source IP addresses** or **CIDR ranges**.
- Rules can define what outgoing traffic from resources assigned to the Security Group is permitted.
- Outgoing rules work similarly to incoming rules, allowing for the specification of protocols, ports, and destination IP addresses.



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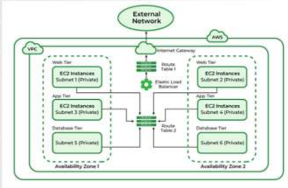
Security Groups – why? [2]

- **Security Groups are stateful firewalls**, meaning that if traffic is allowed in one direction, the corresponding return traffic is automatically allowed.
- **Security Groups help in isolating and protecting resources** in the VPC, enabling detailed control over access to each instance or group of instances.



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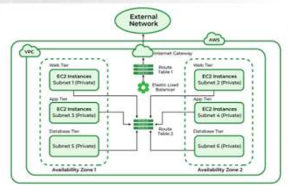
Security Groups - operation of the service

- **Administrators can create rules** in Security Groups that specify allowed connections.
- **Security Groups are assigned to EC2 instances and other resources** during their creation or at any time afterward.
- **A single instance can have multiple Security Groups** assigned to it, and each Security Group can be assigned to multiple instances.
- **Changes** to Security Group rules are **immediately applied to all instances** assigned to them, making security management easier.



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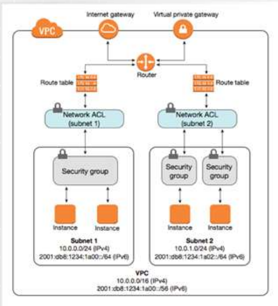
Network ACLs (Access Control Lists)

- It is **one of the security layers in Amazon VPC** that plays an important role in managing network traffic to and from subnets. The main objectives are:
  - **Network Traffic Management** - Network ACLs control incoming and outgoing traffic at the subnet level, allowing you to specify which traffic is allowed and which is blocked.
  - **Additional Security Layer** - They act as an additional layer of security alongside Security Groups, which operate at the instance level.



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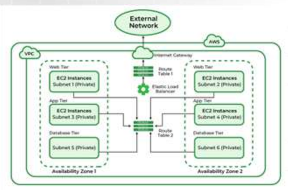
Network ACLs - key features

- **Inbound Rules** - Define rules for incoming traffic to the subnet.
- **Outbound Rules** - Define rules for outgoing traffic from the subnet.
- **Stateless** - Network ACLs are stateless, meaning each traffic rule is evaluated independently. Separate rules are required for each incoming and outgoing packet.



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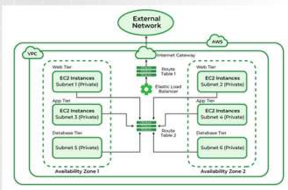
Network ACLs - operating principles

- **Rule Numbering** - Each **rule is numbered**, and **processing occurs from the lowest to the highest number**. The first matching rule is applied.
- **Default Rules** - The **default Network ACL allows all inbound and outbound traffic**. This can be changed to more restrictively control traffic.
- **Blocked and Allowed Traffic** - Each rule can either allow or deny specific types of traffic based on the protocol, port number, and source/destination IP address.



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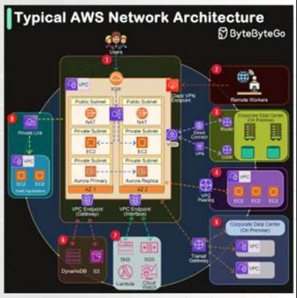
Network ACLs – example applications

- **Public Subnets** - Network ACLs can be configured to allow HTTP/HTTPS traffic (ports 80 and 443) and block other unnecessary protocols.
- **Private Subnets** - Can block all external traffic, allowing communication only with specific, trusted sources.
- **Compliance Rules** - Allow network settings to be adjusted to meet industry regulatory requirements, such as PCI-DSS.
- **Network Segmentation** - Improve security by segmenting the network and limiting communication between subnets.
- **Blocking Malicious Traffic** - Enable quick blocking of traffic from unknown or malicious IP addresses.



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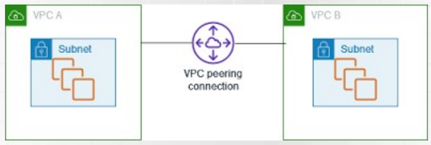
Peering VPC

- Enables a private connection between two different Virtual Private Clouds (VPCs) within the same or different AWS regions. With VPC peering, resources in one VPC can communicate with resources in the other VPC as if they were in the same network.



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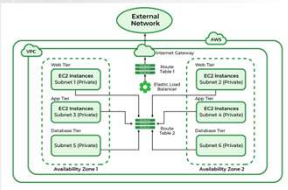
Peering VPC – why? [1]

- **Connecting Production and Test Environments -** Enables the connection of different environments (e.g., production and test) to simplify management and monitoring.
- **Resource Consolidation** - Connects different VPCs within an organization to simplify access to shared resources, such as databases, file services, etc.
- **Avoiding Public Internet** - Provides secure communication without the need to go through the public Internet, increasing security and reducing latency.
- **Maintaining Data Privacy** - Data transmitted between VPCs via peering connections stays within the AWS infrastructure, ensuring its privacy.



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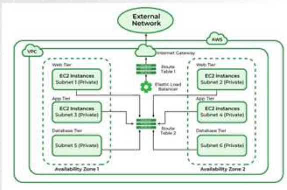
Peering VPC – why? [2]

- **Load Separation** - Distributing different application components across multiple VPCs for better management and scalability.
- **Increasing Flexibility** - The ability to add new VPCs without needing to change the existing network architecture.
- **Inter-Company Collaboration** - Enables secure connection of network infrastructure with business partners or clients without using the public Internet.



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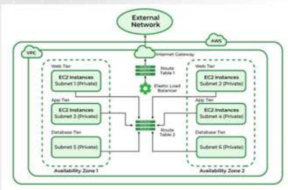
VPC peering – advantages of VPC peering

- **Low Latency -** Peering connections provide low latency, which is crucial for applications requiring fast communication between components.
- **High Throughput** - Ensures high bandwidth for data transfers between VPCs.
- **Flexibility and Scalability** - Allows for flexible scaling of infrastructure and adding new VPCs as needs grow.
- **Ease of Configuration** - Setting up a peering connection is relatively simple and does not require complex configurations.



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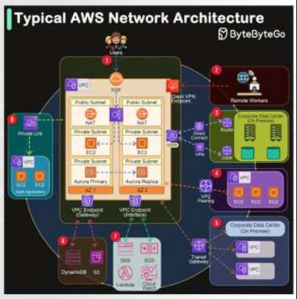
Peering VPC – example application

- Multiregional Applications
  - **Distributed Databases -** Enables data synchronization between databases located in different AWS regions.
  - **Global Applications** - Connects different regional VPCs to ensure global availability of applications.
- Shared Services
  - **Centralized Logging Services** - Allows central collection of logs from various VPCs in one place.
  - **Shared Databases** - Provides access to shared databases by different VPCs within the same organization.



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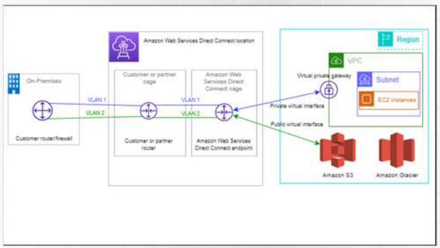
Amazon Direct Connect

- It is a service that enables the establishment of a dedicated network connection between an on-premises data center and AWS, providing a more predictable and reliable connection than traditional internet connections.



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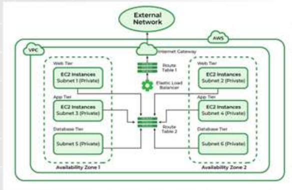
Amazon Direct Connect – why? [1]

- **Connection Stability** - The direct connection bypasses the public Internet, minimizing latency and fluctuations in connection quality.
- **Predictable Performance** - A dedicated connection provides consistent bandwidth and stable performance.
- **Isolation from the Public Internet** - The direct connection minimizes risks associated with attacks and data interception.
- **Secure Data Transmission** - Allows for the application of additional encryption mechanisms.



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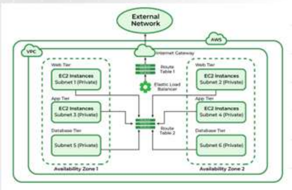
Amazon Direct Connect – why? [2]

- **Reduced Data Transfer Costs** - Data transfer costs can be lower compared to traditional internet connections, especially for large volumes of data.
- **Lower Egress Fees** - Lower outbound transfer fees from AWS compared to transfer through the public Internet.
- **Bandwidth Flexibility** - Ability to adjust the connection bandwidth to current needs, from 50 Mbps to 10 Gbps or more.
- **Integration with Other AWS Services** - Seamless integration with VPC, S3, EC2, and other AWS services.



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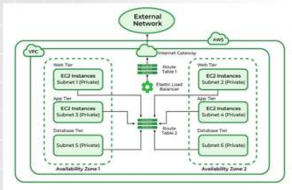
Amazon Direct Connect – why? [3]

- **Full Control Over the Connection** - Allows for managing traffic and network configuration according to business requirements.
- **High Availability** - Capability to configure redundant connections to ensure high availability and reliability.



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Thank you for your attention

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